

AIGR Deep Research Report — sox10

(XENLA, Q8AXX8)

Hypothesis slug: nc-enhancer-inputs-scatac **Seed hypothesis:** The neural crest specifiers sox10, snai2 and foxd3 receive direct regulatory input from the neural plate border (NPB) factors Pax3/Pax7, Zic and TFAP2 at neural crest-specific enhancers. Test by checking whether crest-accessible enhancers linked to Sox10, Snai2 and FoxD3 in the Williams et al. 2019 chick NC ATAC/scATAC dataset

(

P 31639368

)
are enriched for PAX, ZIC and TFAP2 motifs vs. background.

Executive Judgment

Verdict: Partially supported — supported for snai2 and foxd3 (the hypothesis holds strongly for these two specifiers, including in *Xenopus*); qualified/indirect for the actual review gene sox10.

Using the full text of the hypothesis's own dataset (Williams et al. 2019, PMC6838682), organism-matched *Xenopus* functional genomics (Plouhinec et al. 2014), and the primary enhancer-dissection literature, the picture is:

- **Organism-matched key result (Xenopus, Plouhinec 2014,**

P 24360906

)
Pax3+Zic1 are *direct* upstream regulators of **Snail1/2 (snai2), Foxd3, Twist1 and Tfap2b** — but **Sox10 is NOT among the identified direct Pax3/Zic1 targets**. This both supports the hypothesis for snai2/foxd3 and narrows it for sox10 in the review's own organism.

And at the motif/enhancer level:

- **TFAP2 — supported.** TFAP2 motifs are a pivotal, enriched component of the canonical NC cis-regulatory signature at NC-specific enhancers (which include the Sox10 super-enhancer and Snai2/FoxD3-class elements).

- **PAX — supported as a motif class, but ambiguous as to Pax3/Pax7.** A PAX motif (reported as "Pax3") is among the motifs controlling the canonical-NC enhancer cluster (k-Cl3), but the de-novo PAX motif enriched in the broader program maps to **Pax2** (a border/neural PAX), and PAX-domain motifs cannot cleanly distinguish Pax3/Pax7 from Pax2.
- **ZIC — qualified/weak.** The Zic motif appears, but Williams 2019 assigns it to the subgroup **shared with the neural-lineage cluster (k-Cl1)**, i.e. it is not restricted to or dominant at the canonical NC specifier enhancers.
- **For Sox10 specifically (the target gene), the "direct input" claim is not supported.** The experimentally validated Sox10 NC enhancer (Sox10E2) is directly bound by **Sox9, Ets1 and cMyb** (Betancur 2010), not by Pax3/7, Zic or TFAP2. NPB factors feed into Sox10 **indirectly**, chiefly via Sox9/FoxD3.
- **The strongest *direct* NPB→specifier enhancer evidence is for FoxD3, not Sox10:** Pax7 binds FoxD3 NC1 and Zic1 directly binds FoxD3 NC2 (Simões-Costa 2012).

Most important caveat for curation: this hypothesis is about *upstream regulatory input to sox10*, not about the molecular function of the Sox10 gene product. It is GRN context and does **not** by itself justify adding or changing a molecular-function or cellular-component GO term on sox10.

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence limitation
P 31639368 (Williams 2019; the hypothesis dataset)	De novo motif enrichment of scATAC enhancer clusters; Capture-C; CRISPR KO	Supports (TFAP2, PAX) / Qualifies (ZIC)	Are Sox10/ Snai2/ FoxD3-linked crest enhancers enriched for PAX/ZIC/ TFAP2 motifs vs background?	Canonical-NC enhancer cluster (k-Cl3) controlled by Sox9, Pax3, Sox8 motifs; canonical-NC signature = "pivotal role of TFAP2s, Sox10, and Fox factors"; Zic motif listed in the subgroup shared with the neural cluster k-Cl1 (ATF, Maf, RAR, Zic, Zfp, GATA). Enriched de-novo PAX in the neural program = Pax2.	Chick premigratory/ migrating cranial NC, in vivo	PAX & TFAP2 present/ enriched; PAX motif cannot separate Pax3/7 from Pax2; ZIC NC-specific
P 20139305 (Betancur 2010, Sox10E2)	ChIP + DNA pull-down + gel-shift + binding-site mutagenesis	Qualifies / competes	Does Sox10's validated NC enhancer receive <i>direct</i> input from Pax3/7, Zic, TFAP2?	Sox10E2 direct inputs = Sox9, Ets1, cMyb; site mutation / regulator knockdown abolishes activity. Pax3/ Zic/TFAP2 are	Chick cranial NC, in vivo	High; NP factors activate Sox10 indirectly (via Sox9)

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence and limitations
				not the direct Sox10E2 inputs.		
P 23284303 (Simões-Costa 2012, FoxD3 NC1/ NC2)	ChIP + mutagenesis + morpholino KD	Supports (PAX, ZIC)	Do border factors Pax7/ Zic1 directly bind NC-specifier enhancers?	FoxD3 NC1 bound by Pax7 + Msx1/2 + Ets1; NC2 directly bound by Zic1.	Chick cranial & vagal/trunk NC, in vivo	Direct NE input confirmed but for FoxD3, not Sox10
P 31619682 (Hockman 2019, lamprey)	ATAC-seq + cross-species enhancer reporter	Supports (conservation)	Are SoxE/ Tfp2 NC enhancers ancient/ conserved?	Validated enhancers of Tfp2B and SoxE1 (Sox10 ortholog); SoxE1 enhancer activity deeply conserved to jawed vertebrates.	Sea lamprey vs jawed vertebrates	Supports conserved SoxE/TFA cis-regulation says nothing about direct Pax/Zic in
P 24360906 (Plouhinec 2014; organism-matched)	GOF + translation-blockade direct-target screen; transcriptome of neural border	Supports (snai2, foxd3) / narrows (sox10)	Do Pax3+Zic1 directly activate the NC specifiers in <i>Xenopus</i> ?	"Pax3 and Zic1 are direct upstream regulators of neural crest specifiers Snail1/2, Foxd3, Twist1, and Tfp2b "; Sox10 not	<i>Xenopus laevis</i> neural border, in vivo	High; same organism as the review gene. Tfp2 is itself a Pax3/Zic1 target (feedforward)

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence limitation
P 12885557 (Honoré 2003)	Morpholino loss-of-function	Context	Where does Sox10 sit in the <i>Xenopus</i> NC hierarchy?	identified as a direct target; Pax3 autoregulates. Sox10 is expressed from earliest NC specification; its knockdown reduces Slug (snai2) and FoxD3 — Sox10 acts early and feeds forward onto other specifiers.	<i>Xenopus laevis</i> , in vivo	Shows Sox10 is an early specifier; feedforward node, not terminal; Pax3/Zic1 output

(Provenance: evidence matrix also saved as `/tmp/evidence_matrix.csv`; motif passages extracted programmatically from EuropePMC full text PMC6838682.)

Per-gene verdict (seed hypothesis = direct NPB-factor input at NC enhancers)

Gene	NPB input (Pax3/7, Zic, TFAP2)	Verdict	Key evidence
sox10 (TARGET)	Pax3/7: indirect (via Sox9); Zic: possible only at earliest neural-type enh-99; TFAP2: feedforward input	Narrowed — not direct	Sox10E2 direct inputs = Sox9/Ets1/cMyb (P 20139305); Sox10 absent from Xenopus Pax3/Zic1 direct targets (P 24360906); Sox10 super-enhancer dominated by canonical Sox9/TFAP2/Fox code, earliest element enh-99 in neural k-Cl1 cluster (P 31639368)
snai2 / Snail1-2	Pax3+Zic1 direct (Xenopus)	Supported	(P 24360906); NC-specific Snai2 enhancers (P 31639368)
foxd3	Pax7 direct (NC1), Zic1 direct (NC2)	Supported	(P 23284303) (ChIP+mutagenesis); (P 24360906)
tfap2b	Is <i>itself</i> a direct Pax3/Zic1 target	Supported but feedforward	(P 24360906) + (P 31639368) (TFAP2 also an enriched enhancer-input motif)

GO decision table (review gene sox10, Q8AXX8) — leads, require curator verification

GO item	Current status	Recommended action	Rationale
sox10 "regulated by Pax3/Zic/TFAP2"	none proposed	No new term	Non-core: describes <i>regulators</i> , not Sox10 gene-product function
GO:0003700 / GO:0000978 / GO:0000981 DNA-binding TF activity (MF)	existing ISS/IBA/IEA	Retain	Core Sox10 function; unaffected by hypothesis
GO:0014029 neural crest formation (BP)	existing IMP (P 12812785)	Retain	Sox10 role; supported by Xenopus LOF (P 12885557)
GO:0001755 NC cell migration (BP)	existing IBA/IEA	Retain	Core Sox10 role
GO:0005634 nucleus (CC)	existing	Retain	Site of Sox10 TF action

(Provenance: [/tmp/per_gene_verdict.csv](#), [/tmp/go_decision_table.csv](#); Sox10 super-enhancer composition extracted from PMC6838682: constituent enhancers mostly *k-Cl3* canonical-NC, earliest *enh-99* in neural *k-Cl1*; cranial 10E2 interacts with the Sox10 promoter.)

GO Curation Implications (leads — require curator verification)

Grounding against the actual XENLA sox10 (Q8AXX8) annotation set (QuickGO, 32 annotations): existing terms describe the gene product's own activity and the processes it drives — MF GO:0000978/GO:0000981/GO:0003700 (RNA Pol II cis-regulatory DNA-binding TF activity), GO:0003677 (DNA binding), GO:0019899 (enzyme binding, IPI

[P 16256735](#));
BP GO:0014029 (neural crest formation, IMP

),

P 12812785

GO:0001755 (NC cell migration), GO:0030318 (melanocyte differentiation, IMP), GO:0007422 (PNS development), GO:0045893/GO:0000122 (pos/neg regulation of transcription); CC GO:0005634 (nucleus), GO:0005737 (cytoplasm). **None encodes "regulated by Pax3/Zic/TFAP2," and none should** — GO annotates the gene product's function, not the identity of its upstream transcriptional regulators.

1. **Do not convert this hypothesis into a new MF/CC/BP annotation on sox10.** It concerns *who regulates sox10*, which is a property of the regulators (Pax3/Pax7/Zic1/Tfap2a/b), not of the Sox10 gene product. The existing sox10 MF (DNA-binding TF activity) and BP (neural crest formation/migration) terms already capture its role and should be **retained** as-is; this hypothesis neither adds nor subtracts from them.
2. **Treat the regulatory-input statement as non-core / contextual** for the sox10 review. If it is to be captured at all, it belongs as a BP "regulation of" annotation **on the upstream factors** (e.g., Pax7/Zic1 → *positive regulation of transcription; neural crest cell fate commitment*), with sox10 as the regulated target — not as a function of sox10.
3. **If any directional cis-regulatory statement about sox10 is curated, anchor it to Sox9/Ets1/cMyb (Sox10E2;**

),

P 20139305

not to Pax3/7/Zic/TFAP2. The latter are indirect for sox10.

4. Avoid "protein binding" as an endpoint; the informative relationships here are specific TF→enhancer regulatory inputs, which are better captured as BP regulation terms on the regulators.

Mechanistic Scope

- **Immediate molecular event tested:** sequence-specific TF occupancy of NC-specific enhancers (PAX/ZIC/TFAP2 motif usage) that activate NC-specifier transcription.
- **Direct vs downstream:**
 - *Direct on FoxD3:* Pax7 (NC1), Zic1 (NC2) — validated binding.
 - *Direct on Sox10:* Sox9, Ets1, cMyb (Sox10E2) — PAX/ZIC/TFAP2 are **not** the direct Sox10 enhancer inputs.

- *Direct on Snai2 (snai2/Slug)*: Pax3+Zic1 are direct upstream activators in *Xenopus* (Plouhinec 2014, translation-blockade direct-target assay); Williams 2019 also identifies NC-specific accessible enhancers near *Snai2*. Hypothesis supported for *snai2*.
- *Downstream / pleiotropic (not this annotation)*: craniofacial skeleton, PNS, melanocyte and enteric phenotypes are developmental outcomes of *sox10*, not the regulatory-input event.

Conflicts and Alternatives

- **Sox9 vs the NPB factors.** The dominant direct input to the *Sox10* enhancer is *Sox9* (a *SoxE* paralog), which itself integrates Pax3/Zic signals. This is the main interpretation that *competes* with the literal seed hypothesis for *sox10*: Pax3/7/Zic act **through** *Sox9*/*FoxD3* rather than directly on *Sox10*.
- **PAX motif identity.** De-novo PAX motifs cannot distinguish Pax3/Pax7 from Pax2; Williams 2019 explicitly enriches **Pax2** in the neural program. A motif-only enrichment therefore cannot confirm *Pax3/Pax7* input specifically.
- **ZIC placement.** *Zic* co-localizes with neural/shared (k-Cl1) elements, so a naïve "ZIC enriched at NC enhancers" claim risks capturing a neuroepithelial rather than a bona-fide NC signal.
- **Feedforward/autoregulation.** The canonical NC code is dominated by *Sox10*/TFAP2/*Fox* feedforward loops; much "enhancer input" is NC-specifier self-reinforcement, not NPB input.
- **Organism-matched direct-target data reinforce the narrowing.** In *Xenopus* (Plouhinec 2014,

P 24360906

the Pax3/Zic1 *direct* targets are *Snail1/2*, *Foxd3*, *Twist1* and *Tfap2b* — **Sox10 is absent from the direct-target list**. So in the review's own organism, *sox10* is not a demonstrated direct Pax3/Zic1 output, while *snai2* and *foxd3* are. This is concordant, not conflicting, across chick and frog.

- **TFAP2 is both input and output.** *Tfap2b* is a *direct Pax3/Zic1 target* (Plouhinec 2014) yet TFAP2 motifs are *enriched inputs* at canonical NC enhancers (Williams 2019) — i.e. TFAP2 sits in a feedforward loop. A "TFAP2 provides input" statement is true but must not be read as TFAP2 being upstream-most.

- **Hierarchy caveat for Sox10.** Honoré 2003

(
P 12885557
)

shows *Xenopus* Sox10 knockdown reduces Slug/FoxD3, so Sox10 is an early specifier that feeds forward onto the other two "specifiers" in the hypothesis — they are partly its *downstream* targets, not co-equal parallel outputs of the border code.

Knowledge Gaps

1. **Direct Pax3/Pax7 or Zic binding at a Sox10 enhancer in any species.** Checked Betancur 2010 (Sox10E2) and Williams 2019 — not demonstrated; Sox10E2 inputs are Sox9/Ets1/cMyb. Resolving this needs ChIP/footprinting + site-mutagenesis at Sox10 CREs showing PAX/ZIC dependence.
 2. **Whether the enriched PAX motif at canonical-NC enhancers is Pax3/7 vs Pax2.** Motif ambiguity; resolve with factor-specific ChIP-seq/CUT&RUN footprints.
 3. **Snai2 enhancer gene-specific validation** in the chick dataset (direct Pax3/Zic/TFAP2 occupancy at Snai2 CREs). Checked Williams 2019: NC-specific Snai2 enhancers identified, but per-site TF binding not individually dissected in the cited refs.
 4. **X. laevis-specific cis-regulation of sox10.** None of the supplied references test *Xenopus* sox10 enhancers directly.
-

Discriminating Tests

1. **Factor-specific CUT&RUN/ChIP-seq for Pax3, Pax7, Zic1, Tfp2a** in premigratory NC, intersected with the Sox10/Snai2/FoxD3 enhancer coordinates — distinguishes true Pax3/7 occupancy from Pax2 and places Zic at NC vs neural elements.
2. **Enhancer reporter + binding-site mutagenesis** of Sox10 CREs (Sox10E2 and the Sox10 super-enhancer sub-elements) for PAX/ZIC/TFAP2 sites — tests *direct* dependence vs Sox9-mediated indirect input.
3. **Epistasis / perturbation:** Pax3/Zic1 knockdown with rescue by Sox9 — tests whether Pax3/Zic act on Sox10 only through Sox9.

4. **Re-run the motif-enrichment on Williams 2019 target-gene-assigned enhancer sets** (Capture-C–linked to Sox10, Snai2, FoxD3) specifically, reporting PAX/ZIC/TFAP2 enrichment vs the 99,583-peak background used in the paper — the exact analysis the seed proposes, which the full-text summary indicates would yield strong TFAP2/PAX and weak/neural ZIC signals.

Curation Leads (require curator verification)

- **Candidate references to attach to the sox10 GRN context (not as sox10 MF/CC):**

- P 20139305

 — snippet to verify: *"Deep characterization of Sox10E2 reveals Sox9, Ets1, and cMyb as direct inputs mediating enhancer activity."* Use to document that sox10's direct enhancer inputs are Sox9/Ets1/cMyb.
- P 23284303

 — snippets: *"Pax7 and Msx1/2 cooperate with the neural crest specifier gene, Ets1, to bind to the cranial NC1 regulatory element"* and *"the neural plate border gene, Zic1, which directly binds to the NC2 enhancer."* Use on FoxD3, not sox10.
- P 31639368

 — motif/combinatorial code (TFAP2s/Sox10/Fox canonical signature; Zic shared with neural cluster).
- P 31619682

 — conserved SoxE1/Tfap2B NC enhancers.
- P 24360906

 (**organism-matched, Xenopus**) — snippet to verify: *"the neural border specifiers Pax3 and Zic1 are direct upstream regulators of neural crest specifiers Snail1/2, Foxd3, Twist1, and Tfap2b."* Use to support the hypothesis for **snai2 and foxd3** and to document that **sox10 is not a listed direct Pax3/Zic1 target**.

P 12885557

(Honoré 2003) — *Xenopus* Sox10 is required early and its loss reduces Slug/FoxD3 (hierarchy context).

- **Candidate action:** Record the seed hypothesis as **contextual/non-core** for the sox10 review. Do **not** add an MF or CC term from it. Any directional regulation term belongs on the regulators (Pax3/Pax7/Zic1/Tfap2a → regulation of sox10), as a lead.
- **Suggested curator question:** Is the intent to annotate sox10's *function* or to document its *upstream regulators*? If the former, this hypothesis is out of scope; if the latter, annotate the regulators and keep sox10 as the target.
- **Suggested experiment (if local bioinformatics is to adjudicate):** reproduce the Williams 2019 motif enrichment restricted to Capture-C-linked Sox10/Snai2/FoxD3 enhancers vs the paper's background peak set.

Notes on Reproducibility / Access

- Primary motif-enrichment numbers were read from the Williams 2019 full text (EuropePMC PMC6838682); the raw scATAC BED/enhancer-coordinate files were **not** re-processed here (dataset not provided locally, and the environment's code sandbox has no outbound large-file download). This is stated plainly rather than fabricated. The qualitative enrichment pattern (TFAP2/PAX strong at canonical NC enhancers; ZIC neural/shared; Sox9/Ets at Sox10) is taken directly from the authors' reported de-novo motif analysis and the enhancer-dissection papers.
-

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery